MULTI RETRO DRAM TESTER

Comprehensive User Manual & Technical Reference
Based on Firmware V4.13|Multi Retro Dram Commander v5.31
Download the latest Multi Retro Dram PC Software which can be found at: -

https://multiretrodramtester.co.uk/updates.html

Any issues or queries etc. please email me - retrodramtester@gmail.com

⚠️ **IMPORTANT – READ BEFORE USE**

The Multi Retro DRAM Tester can be powered from either the USB-C port OR via a 12 V DC center-positive power supply.

ONLY CONNECT TO ONE POWER SOURCE AT A TIME!

DO NOT connect USB-C and 12 V DC simultaneously

1. Overview

The Multi Retro DRAM Tester is a specialist diagnostic and characterisation instrument designed for testing vintage Dynamic RAM (DRAM) devices from the late 1970s through to the 1990s. Unlike simple pass/fail checkers, this tester executes industry-standard memory algorithms using precisely controlled timing and voltage conditions.

The platform is built around a Dual-core ARM Cortex-M0+ processor, allowing control of RAS, CAS, WE, address, and data lines while supporting both single-supply and multi-rail DRAM.

2. Power & Electrical Characteristics

There are 2 onboard DC-DC converters create the necessary voltages for the Multi-Rail DRAM Socket – SK2 (+12V & -5V)

2.1 Power Inputs (Only connect to one power source!)

Input	Function
USB-C	Bench operation or PC-controlled mode
12 V DC (center-positive, 2.1 mm)	Bench operation or PC-controlled mode

3. ZIF Sockets, Voltage & Orientation

3.1 Socket Architecture

Two ZIF sockets are provided to safely support different DRAM voltage requirements. An LED next to each socket indicates the correct socket for the currently selected DRAM.

Power to the selected DRAM chip is auto routed, there are no switches to set unlike some Arduino based DRAM testers.

ZIF Socket SK1 — Single-Supply +5 V DRAM

- HM4816 (16K × 1, a 5V-only 4116 derivative)
- OKI M3732L (32K × 1)
- OKI M3732H (32K × 1)
- TMS4532-NL3 (32K × 1)
- TMS4532-NL4 (32K × 1)
- 4164 (64K × 1)
- 4128 (128K × 1 - Piggyback)
- 41256 (256K × 1)
- 4416 (16K × 4)
- 4464 (64K × 4)
- 411000 (1M × 1)
- 44256 (256K × 4)
- 514400 (1M × 4)

ZIF Socket SK2 — Multi-Rail DRAM (−5 V / +5 V / +12 V)

- 4116 (16K × 1)
- MK4027 (4K × 1)
- TMS4108 (8K × 1)

3.2 Pin-1 Orientation

Pin-1 orientation is critical. The DRAM Pin-1 marker must align with the PCB silkscreen marking. Incorrect orientation may damage the Tester / DRAM and invalidate test results.

4. Operating Modes

4.1 Standalone Mode

The tester can operate independently using the onboard display and rotary encoder, making it suitable for bench testing and field diagnostics.

4.2 PC-Controlled Mode

Using the Multi Retro Dram Commander Windows application, the tester can be fully controlled over USB serial (115200 baud). This mode exposes advanced configuration, logging, and automation features.

5. DRAM Test Algorithms

March B

A fast algorithm effective at detecting stuck-at faults and basic coupling issues.

March C- (Minus)

A comprehensive industry-standard DRAM test capable of detecting stuck-at, transition, coupling, and address decoder faults. This is the recommended default algorithm.

March C- Mix
Executes March C- using both standard access timing and Fast Page Mode, exposing marginal DRAM that passes at low speed but fails under tighter timing constraints.

Checkerboard
Pattern-sensitive test using alternating bit patterns to reveal adjacent-cell interference.

True Row Retention
Implements a genuine per-row retention test by writing, delaying without refresh, and verifying each DRAM row independently.

Extreme / Stress
A user-configurable stress mode that combines multiple algorithms into a single extended test run.

6. Measured Test Timings (Using Firmware revision V4.08b)

DRAM	Organisation	March B	March C-	March B + Retention	Retention Time
4027	4K × 1	39 ms	57 ms	372 ms	2 ms
4108	8K × 1	67 ms	101 ms	666 ms	2 ms
4116	16K × 1	120 ms	191 ms	740 ms	2 ms
4816	16K × 1	120 ms	191 ms	740 ms	2 ms
4532	32K × 1	248 ms	402 ms	1.43 s	2 ms
3732	32K × 1	248 ms	402 ms	1.43 s	2 ms
4164	64K × 1	443 ms	727 ms	2.7 s	4 ms
4128	128K × 1	1.25 s	2.07 s	3.8 s	4 ms
41256	256K × 1	1.96 s	3.25 s	7.8 s	4 ms
4416	16K × 4	197 ms	320 ms	2.4 s	4 ms
4464	64K × 4	750 ms	1.25 s	3.7 s	4 ms
44256	256K × 4	3 s	5 s	10.5 s	4 ms
411000	1M × 1	7.4 s	12.2 s	30 s	8 ms
514400	1M × 4	12 s	20 s	58 s	16 ms

7. Automatic Pre-Tests (These settings are best left on)
Before executing the main algorithm, the tester performs a series of automatic validation steps:
- Pin Check – detects shorts and stuck pins before power is applied
- Wake-Up Refresh – stabilises DRAM prior to testing
- Presence Check – detects whether a DRAM is fitted
- Address Sweep – verifies address line integrity
- Data / WE Check – validates data and write-enable operation

8. Settings Menu – Complete Reference
The Settings menu allows control over test execution. All settings are stored in non-volatile memory and restored on boot, except Row Retention time, which is intentionally volatile.

Test
Selects the DRAM test algorithm to execute.

Access Mode
Chooses Standard Page or Fast Page access.

Loop Count
Defines how many times the test repeats. 0 = infinite loop.

On Error
Stop immediately on failure or **Restart** the next loop.

EOT (End Of Test) – This setting is for loop mode.
Keeps DRAM powered or Cycles power between loops.

Phase Messages
Displays internal algorithm phase information. Disable for maximum speed.

Pre-Tests
Enable or disable automatic safety and validation checks (These are best to be left ON!)

Result Size
Small = detailed failure data, Large = prominent PASS/FAIL indication.

Retention Time
Adjusts delay for Row Retention testing. Increasing this can extend the test time. (Debug only!)

Cycle Delay
Delay inserted between loops when looping is enabled. (Best left at a minimum of 500ms!)

Extreme / Stress Config
Selects which algorithms are included in Extreme mode. (Combination of tests)

Buzzer Settings (Pass & Fail SFX can be individually turned On/Off)
Configure pass/fail sounds, tones, SFX and enable states.

Firmware Update (See bottom of page for firmware update instructions)
Scannable QR code link to the update page showing updates for the F/W, S/W & Manuals.

Reset Defaults
Restores all persistent settings to factory defaults.

Test Hardware (Manual or Auto)
Exercises all 20 of the signals on SK1 for hardware verification – P01 (Pin1) to P19 (Pin19)

9. Settings at a Glance

Setting	Purpose	Options	Persisted
Test	Algorithm selection	March / Checkerboard / Retention / Extreme	Yes
Access Mode	Timing mode	Std / Fast	Yes
Loop Count	Repeat count	1 / Fixed / Infinite	Yes
On Error	Failure handling	Stop / Restart	Yes
EOT	Power handling	On / Cycle	Yes
Phase Messages	Show phases	On / Off	Yes
Pre-Tests	Safety checks	Pin / Presence / Addr / Data	Yes
Result Size	Display style	Small / Large	Yes
Retention Time	Retention delay	User adjustable	No
Cycle Delay	Delay between loops	Preset	Yes
Extreme Config	Stress composition	User selectable	Yes
Buzzer Settings	Audio feedback	Enable / Tone select	Yes
Reset Defaults	Factory reset	Confirm	—
Test Hardware	Self test	Manual / Auto	—

10. Typical Test Workflow

1. Apply power using USB-C or 12 V DC (never both).
2. Select the DRAM type.
3. Observe the illuminated socket LED.
4. Insert the DRAM with correct Pin-1 orientation.
5. Select the desired test algorithm.
6. Start the test.
7. Review PASS / FAIL results.

11. Failure Messages

- If a DRAM chip fails a test and you are in LARGE Results Size then the tester will display only a large FAIL message.
- To see a more detailed error message then you need to select Results Size: Small
- Typical Error message March C- (r0, w1) Addr:16 E:0 G:1

  March C-: This is the specific test algorithm running. It is a strict industry-standard test designed to find complex memory faults (stuck bits, coupling faults, address decoder faults).

  (r0, w1): This is the specific **phase** or step of the algorithm where it failed.

  - **r0**: The tester attempted to **Read** a Logic **0** from the memory cell.
  - **w1**: The plan was to immediately **Write** a Logic **1** after reading, but it failed at the read step.

Addr:16: This is the specific memory address where the error occurred. In your tester code, this is displayed in **Hexadecimal**.

- Hex 16 = Decimal 22.
- This is the 23rd cell in the memory array (since it starts at 0).

E:0: **Expected** value. The tester expected to find the data line Low (Logic 0).

G:1: **Got** (Actual) value. The tester actually found the data line High (Logic 1). Expected value was **0** and the Got value was **1**, this specific memory cell is likely suffering from a **"Stuck-At-1" (Stuck High)** fault.

- Address = (Current Row × Total Columns) + Current Column

12. Troubleshooting

- PC shows OFFLINE: verify USB cable and selected COM port.
- Wrong socket LED: confirm DRAM selection.
- Immediate pin failure: inspect orientation and bent pins.

13. Updates & Resources

https://multiretrodramtester.co.uk/updates.html
(Latest firmware / Windows 10/11 PC Software and Operating instructions)

Websites: -
https://multiretrodramtester.co.uk/index.html
https://dramtester.co.uk

14. Firmware update (Windows 10/11)

Download the latest firmware from *https://multiretrodramtester.co.uk/updates.html*

1. Download the **latest firmware (*.UF2)**
2. Use a **USB-C to USB-A or USB-C cable** (depending on your PC USB Ports).
3. Connect the **Multi Retro DRAM Tester** to your PC.

To Enter Bootloader Mode
Use **either** method below: (Do not have a 12V PSU connected!)

- **Method 1**
  Hold the **BOOT** button down → Press and release **RESET**
- **Method 2**
  Hold **BOOT button** while connecting the USB cable

4. A new removable drive will appear named: - "RPI-RP2"
5. Drag and drop the ***.UF2 firmware file** onto this drive.
6. The tester will **automatically program, reboot, and run the new firmware**.